



1501 - A Comparison of Matting Efficiency and COF Using AquaMatte® 22, AquaMatte 26HD, AquaMatte 31 and PropylMatte 31HD

Introduction

AquaMatte 22, AquaMatte 31, and AquaMatte 26HD are a series of micronized oxidized polyethylene waxes designed for gloss reduction and stability in waterborne systems. PropylMatte 31HD is micronized polypropylene that is designed for use as a matting agent.

PropylMatte 31HD and AquaMatte 26HD have been density-modified to 1.07 g/cc to eliminate wax flotation, especially in water based and UV/EB systems.

	AquaMatte 22	AquaMatte 26HD	AquaMatte 31	PropylMatte 31HD
Melting Point (C°)	135-140	135-140	135-140	160-170
Density @ 25°C (g/cc)	0.99	1.07	0.99	1.07
NPIRI Grind	2.0-3.0	3.0-4.0	5.0-6.0	5.0-6.0
Mean Particle Size (µm)	6.0-8.0	6.0-8.5	8.0-12.0	8.0-12.0
Maximum Particle Size (µm)	22.0	26.16	31.0	31.0

Objectives

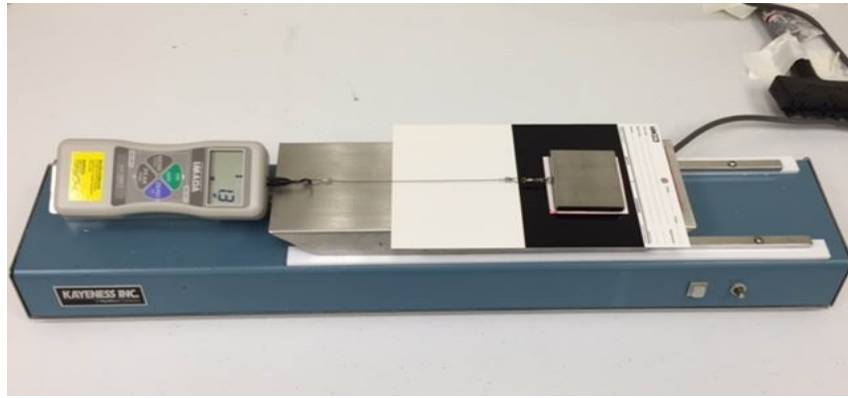
Compare the matting efficiency and coefficient of friction (COF) using AquaMatte 22, AquaMatte 31, AquaMatte 26HD and PropylMatte 31HD.

Study Methodology

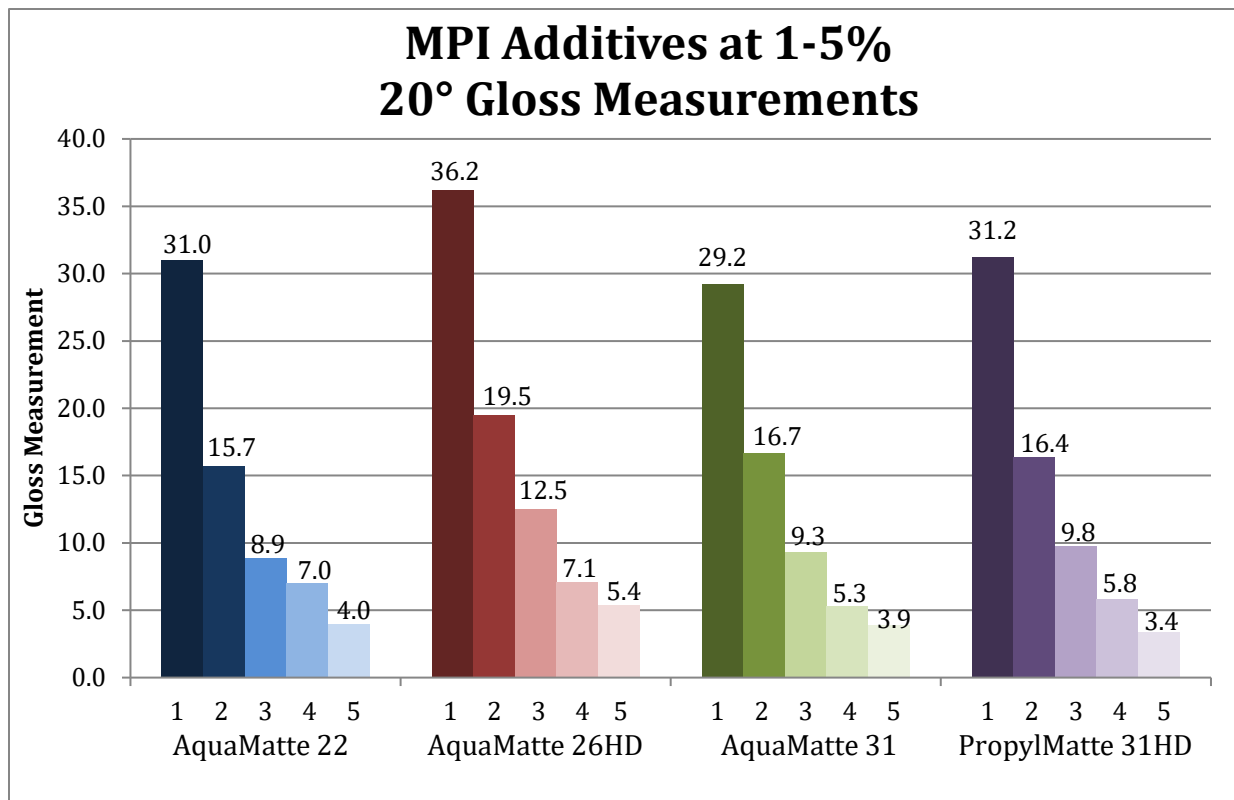
Aqueous polyurethane dispersions were made at a 1%, 2%, 3%, 4% and 5% dosage (based on total formula weight) for each product. The dispersions were drawn down using a 0.003" (76 µm) Bird applicator. Panels were dried for one week before measurements were taken.

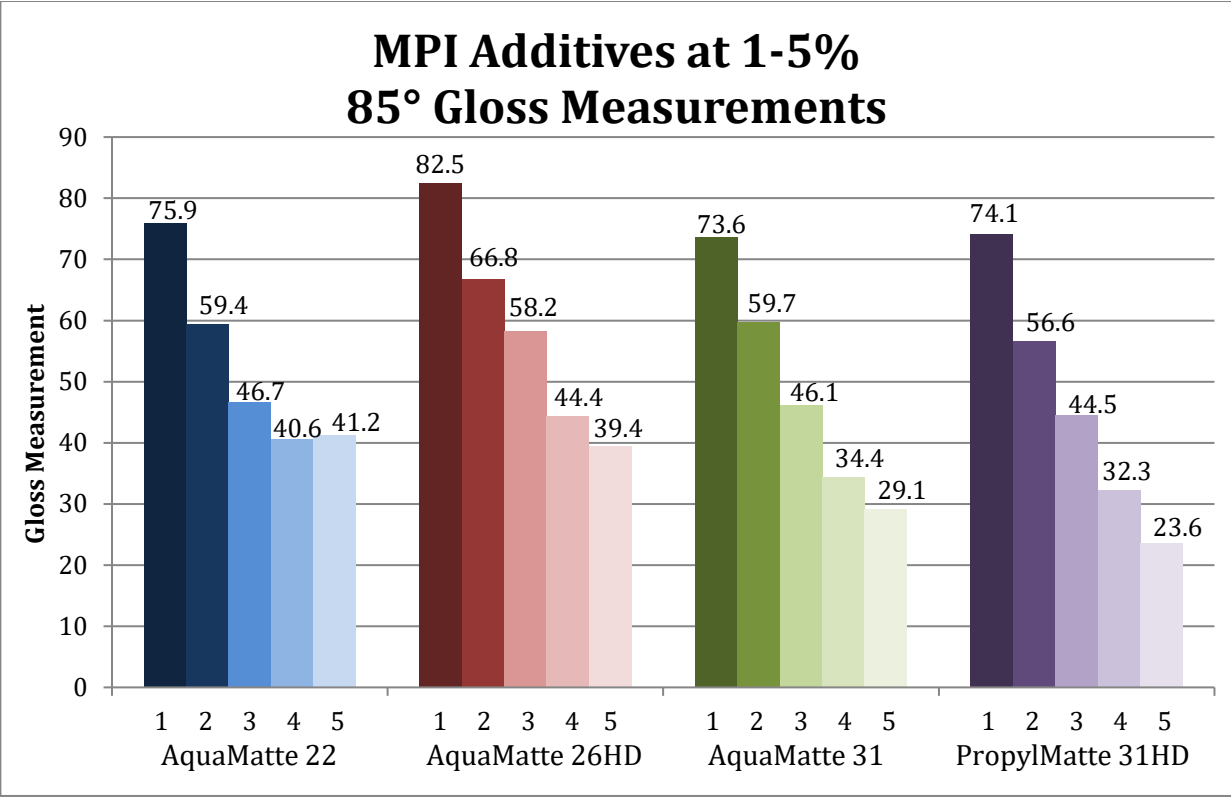
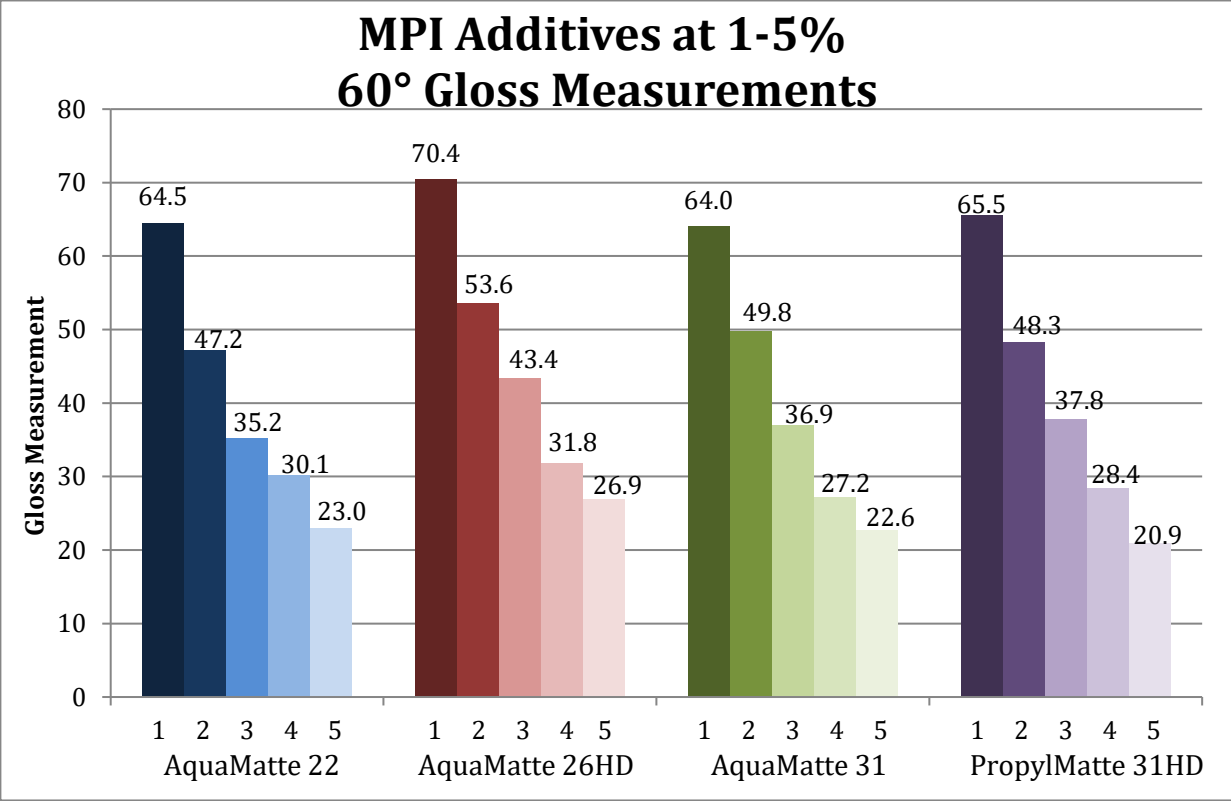
Specular gloss was measured at 20°, 60° and 85° on the black area of a coated lacquered Leneta Chart Form 2A.

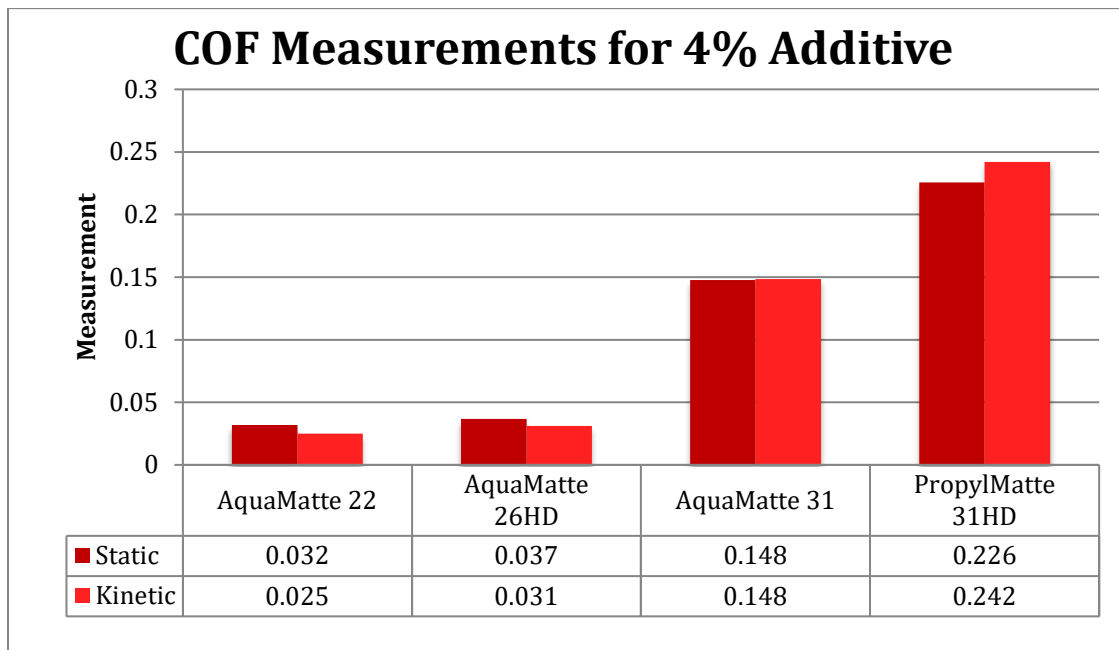
COF was measured using a Kayeness Model D1055 tester. A Leneta card, coated with the cured test material, is secured to the moving platform. A digital force gauge attaches directly to a 200 gram sled which sits on the Leneta card. The cured coating is affixed to the sled. The platform moves at a rate of 0.1 inches per second. The force gauge measures the gram-force load and determines static and kinetic COF.



Results







Conclusions

Matting Efficiency

Polymeric matting agents can provide a range of gloss reduction down to 20 at 60°.

- A *semi-gloss* (45-50 at 60°) effect can be achieved by adding about 2% polymeric wax
- A *satín* effect (20-25 at 60°) can be achieved by adding about 5% of polymeric wax

PropylMatte 31HD provided the best overall matting efficiency at all 3 angles.

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COF

- AquaMatte 22 provided the lowest COF (highest slip), followed closely by AMT-26HD
- PropylMatte 31HD provided the highest COF (lowest slip) as would be expected for a polypropylene-based wax additive

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